

Modern Portfolio Theory and CC Portfolio Construction

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Modern Portfolio Theory Overview

Modern Portfolio Theory is a framework for constructing investment portfolios that aim to maximize expected return for a given level of risk, or equivalently, minimize risk for a given level of expected return.

- **Harry Markowitz**, an American economist, introduced MPT in 1952 through his seminal paper "Portfolio Selection"
- His groundbreaking work earned him the **Nobel Memorial Prize in Economic Sciences in 1990**, shared with William Sharpe and Merton Miller, for their contributions to the development of MPT and the Capital Asset Pricing Model (CAPM).

Uses historical price data to estimate the correlation, variance and returns of differently weighted portfolios – then, will according to the efficient frontier present different portfolio theories with different weightings for different risk appetites.

The fundamental idea is:

- **Diversification** of assets is the safest way to higher returns with less risk.
- Aim is to reduce idiosyncratic risk by holding assets from a variety of "sectors".
 - High correlation between different assets increases the st.d of the portfolio.
 - Yet, holding Mag7 f.ex would have outperformed... risk tolerance dependent.

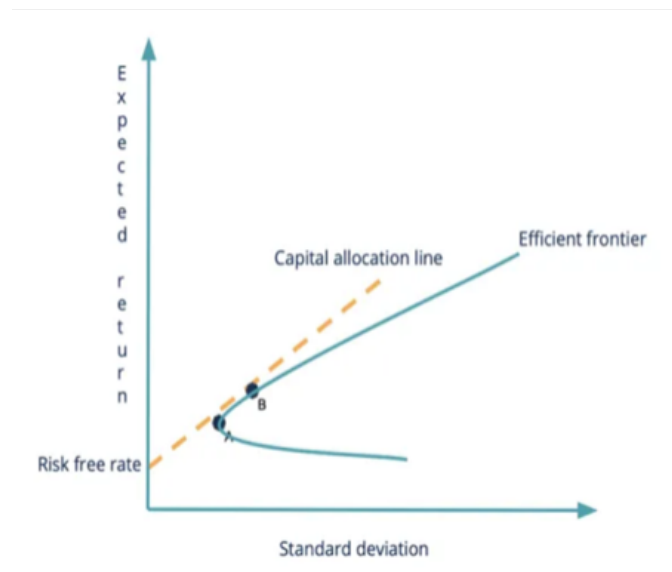
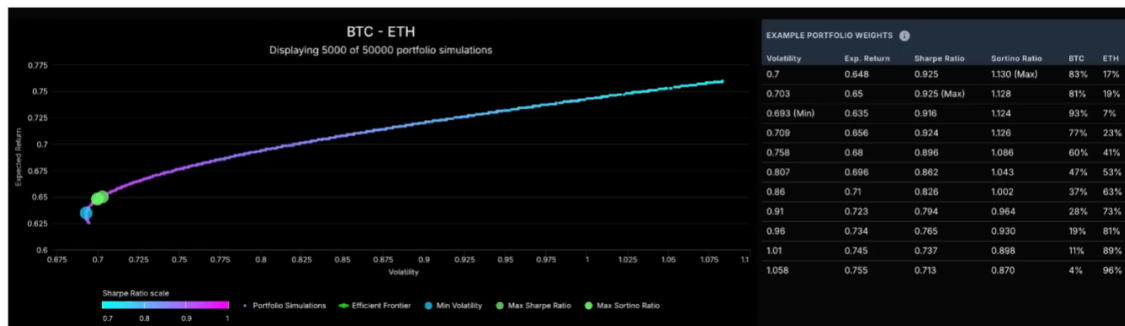
The key MPT assumptions are:

1. Returns are distributed along a normal "bell curve".
2. Correlations do not change over time.
3. Investors will make rational investment decisions and will avoid unnecessary risk.
4. Asset volatility is known and constant.
5. No taxes or transaction costs.

Diversification eliminates Idiosyncratic Risk, not systemic risk

Portfolio Frontier:

According to the Modern Portfolio Theory, a portfolio frontier, also known as an efficient frontier, is a **set of portfolios that maximizes expected returns for each level of standard deviation (risk)**. A typical portfolio frontier is illustrated below:



Expected Return: expected value of the probability distribution of the possible returns it can provide to investors

Volatility: Standard deviation (st.d) of an asset. It is used to determine how widely spread out the asset movements are over time (in terms of value)

Standard Deviation of Portfolio depends on:

- Standard Deviation of each asset (price history).
- The weights of each asset in the portfolio.
- The correlation between all the assets.

Higher Sharpe Ratio → Better Portfolio:

$$\text{Sharpe Ratio} = \frac{R_p - R_f}{\sigma_p}$$

Where:

- R_p = Return of portfolio
- R_f = Risk-Free rate
- σ_p = Standard deviation of portfolio's excess return

Efficient Frontier:

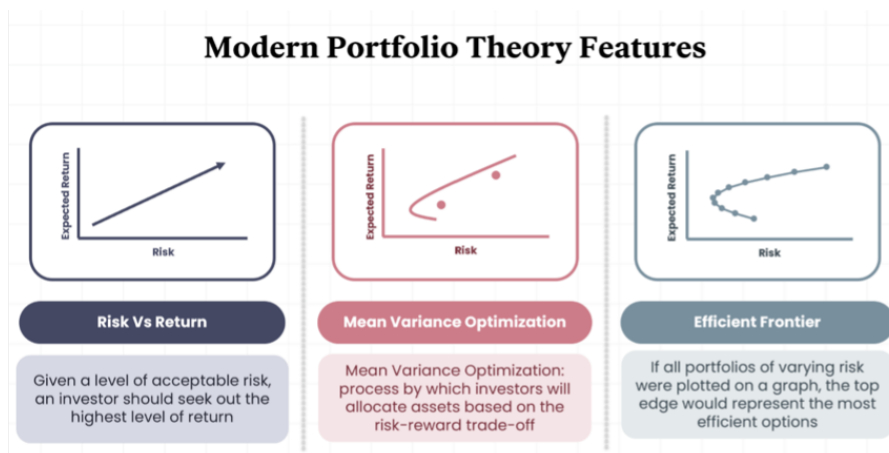
- **Point “A”** on the efficient frontier is the **minimum variance portfolio** the combination of risky-assets that minimizes standard deviation/risk.
- **Point “B”** is the **optimal market portfolio**, which consists of at least one risk-free asset. It is depicted by the line that is tangent to the efficient frontier, which is also called the Capital Allocation Line (CAL).

$$\text{Slope of CML} = \frac{E[R_m] - R_f}{\sigma_m}$$

Capital Allocation Line:

- The line that depicts the risk-reward tradeoff of assets that carry idiosyncratic risk. The slope of **the CAL is called the Sharpe ratio**, which is the increase in expected return per additional unit of standard deviation (reward-to-risk ratio).
- Line changes for each portfolio weighting and provides different optimal portfolios. along efficient frontier (is return of optimal risky portfolio)

Sharpe Ratio is used to find the single optimal portfolio (Max Sharpe). Efficient frontier itself is line that minimizes risk for different given expected returns



4. Example

- Risk-free rate: 2%
- Portfolio expected return: 10%
- Portfolio standard deviation: 15%

$$\text{Sharpe} = \frac{10\% - 2\%}{15\%} = \frac{8}{15} \approx 0.53$$

Interpretation:

For every 1 unit of risk taken, this portfolio delivers 0.53 units of excess return.

- If another portfolio had Sharpe = 0.8, it would be **more efficient**, even if its raw return was lower.

Mean-Variance Optimisation – Wanting a particular targeted level of return whilst

seeking the lowest level of risk: Done through MVO :

The MVO is the process by which investors will allocate portfolios based on the risk-reward trade-off. This is done by investors varying the weightings in various assets or asset classes to determine which portfolios provide the best risk-return trade-off. Such analysis results in what is termed as the “Efficient Frontier”

Mathematical Foundations

Portfolio Return

The expected return of a portfolio is the weighted sum of the expected returns of its individual assets, where:

- $E[R_p] = \text{Expected return of the portfolio.}$
- $w_i = \text{Weight of asset } i \text{ in the portfolio}$
- $E[R_i] = \text{Expected return of asset } i \text{ (CAPM for return or historical model with formula.)}$

Calculated by:

$$E[R_p] = \sum_{i=1}^n w_i E[R_i]$$

Portfolio Variance and Standard Deviation:

The variance of a portfolio’s return accounts for both the variances of individual assets and their covariances, where:

- $(\sigma_p)^2 = \text{Variance of the portfolio's return.}$
- $w_i, w_j = \text{Weights of assets } i \text{ and } j \text{ in the portfolio.}$
- $\sigma_{ij} = \text{Covariance between return of assets } i \text{ and } j \text{ (math formula to obtain this.)}$
- $\text{The standard deviation } \sigma_p, \text{ is the square root of the variance and represents the portfolio's total risk.}$

Calculated by:

$$(\sigma_p)^2 = \sum_{i=1}^n \sum_{j=1}^n w_i w_j \sigma_{ij}$$

Efficient Frontier and Optimization

- The efficient frontier is derived by solving the optimization problem that maximizes expected return for a given level of risk or minimizes risk for a given level of expected return.
- This involves using quadratic programming techniques to find the optimal asset weights that satisfy the desired risk-return profile.

Practical Application in Portfolio Construction Steps in Building an Optimal Portfolio

- **Data Collection:** Gather historical data on asset returns, variances, and covariances.
- **Return Estimation:** Estimate the expected returns for each asset (CAPM or Historical Data).
- **Risk Assessment:** Calculate the variances and covariances to understand the risk profile.
- **Optimization:** Use mathematical optimization techniques to determine the optimal asset weights that maximize return for a given level of risk.
 - Solved using quadratic programming excel or Lagrange multipliers if analytically. Calculus to solve for Max Expected return and min Por Variance.
- **Implementation:** Construct the portfolio by allocating capital according to the optimal weights.
- **Monitoring and Rebalancing:** Regularly review and adjust the portfolio to maintain the desired risk-return profile.

Pros and Cons of MPT

Pros

- Modern Portfolio Theory helps investors manage risk by diversifying their portfolios.
- Modern Portfolio Theory allows investors to maximize their expected return for a given level of risk.

Cons

- While Modern Portfolio Theory is very influential, it is important to be aware of the criticisms:
- Modern Portfolio Theory is too simple. More complex models have been created adding more factors other than risk and return.
- Backward looking analysis can have low predictive value for the future.
- Not all investors act rationally but Modern Portfolio Theory assumes they do.
- Not all investors have the ability to invest in risky assets alone and the markets aren't always efficient.

Things to consider re crypto Portfolio:

Modern Portfolio Theory: Streamlined Steps Step 1: Collect inputs - Expected returns of assets: $\mu = [\mu_1, \mu_2, \dots, \mu_n]^T$ - Covariance matrix of asset returns: $\Sigma = [\text{Cov}(R_i, R_j)]_{n \times n}$ - Risk-free rate: R_f Step 2: Define portfolio return and risk $R_p = w^T \mu$ $\sigma_p = \sqrt{w^T \Sigma w}$ where $w = [w_1, w_2, \dots, w_n]^T$ are the portfolio weights. Step 3: Define Sharpe ratio $S = \frac{R_p - R_f}{\sigma_p} = \frac{w^T (\mu - R_f \mathbf{1})}{\sqrt{w^T \Sigma w}}$	Step 4: Maximize Sharpe ratio - Solve the optimization problem: $\max_w \frac{w^T (\mu - R_f \mathbf{1})}{\sqrt{w^T \Sigma w}}$ - Analytical solution using matrix algebra: $w^* \propto \Sigma^{-1} (\mu - R_f \mathbf{1})$ - Normalize if needed ($\sum_i w_i = 1$). Step 5: Obtain optimal portfolio - Portfolio weights $w^* = \text{max Sharpe / tangency portfolio}$ - Expected return: $R_p^* = w^{*T} \mu$ - Portfolio risk: $\sigma_p^* = \sqrt{w^{*T} \Sigma w^*}$ - Maximum Sharpe ratio: $S_{\max} = \frac{R_p^* - R_f}{\sigma_p^*}$
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- Only relevant CC's are ETH and BTC for appropriate allocation.
- Marginal amount allocated to "ALT's", given downside risk vs. upside is not efficient by MPT standards, when you can have BTC.
- Take into Cash Positions for MPT aswell: 30% recommended.
- Cash is even more relevant when timing with Market Cycle Theory.